

MATSCI 465: Advanced Electron Microscopy & Diffraction

Project Proposal

Project Title: Evaluating Unsupervised Clustering Strategies for Identifying Degradation-Related Chemical Domains in SEM-EDS Spectrum Imaging of Zinc-Soap Affected Paint Layers

Name, Email, Department: Elise Etrheim, eliseetrheim2027@u.northwestern.edu, Northwestern University Department of Materials Science & Engineering

Submission Date: 13 February 2026

Proposal

This project will develop and evaluate a reproducible unsupervised machine-learning workflow for analyzing SEM-EDS elemental maps of a zinc-soap protrusion in a zinc-oxide oil paint test panel. Zinc soaps are a major conservation concern due to their spatially heterogeneous formation and migration within paint layers, leading to cracking, delamination, and surface deformation. Interpretation of EDS maps, however, typically relies on qualitative visual assessment of elemental intensity distributions. As heritage science increasingly produces large, high-dimensional analytical datasets, scalable computational approaches are needed to evaluate whether such data can objectively resolve degradation-related chemical domains. This project investigates whether unsupervised learning can recover spatially coherent regions associated with zinc soap formation and surrounding paint matrix from SEM-EDS data alone, and whether those regions can be chemically interpreted using independent point-based EDS measurements.

The primary objective is to evaluate whether clustering methods can partition EDS data into chemically meaningful, degradation-associated regions while simultaneously developing a modular and reproducible computational pipeline. Each pixel in the EDS map will be represented as a multivariate feature vector of normalized elemental intensities. After background correction and optional spatial smoothing, Principal Component Analysis (PCA) will be used to examine variance structure and reduce dimensionality. K-means clustering will then be applied to identify chemically distinct groups, with cluster quality evaluated using silhouette scores and stability analysis across multiple values of k . In addition, K-means will be compared against an alternative clustering method, such as hierarchical clustering or Gaussian Mixture Models, to assess robustness of segmentation outcomes.

Cluster labels will be reshaped into spatial maps to visualize chemical domains within the protrusion and assess whether soap-rich regions, diffusion fronts, and surrounding paint

matrix can be differentiated. To strengthen chemical interpretation, cluster-mean compositions or representative spectra will be compared against point-based EDS spectra collected from known regions of the sample, including gel-like zinc soap and surrounding paint layers. Success will be evaluated based on cluster stability, spatial coherence, and agreement with known compositional trends reported in the literature. As an extension, clustering results from aerial and cross-sectional EDS datasets will be compared to evaluate whether sample orientation influences the ability to resolve degradation-related chemical domains. The final deliverables include a documented analysis pipeline, PCA visualizations, cluster-validation metrics, spatial cluster maps, and a critical discussion of methodological strengths and limitations.

Dataset Information

Dataset Source:

The dataset is publicly available through the National Institute of Standards and Technology (NIST) Public Data Repository. It accompanies the publication *Characterization of zinc carboxylates in an oil paint test panel* (Romano et al., 2019) and is at:

<https://data.nist.gov/od/id/8C40CFA7931709DAE0532457068179072082> (Accessed 02/11/2026)

Data Type(s):

The dataset contains multimodal analytical measurements collected from a zinc-soap protrusion in a zinc-oxide oil paint test panel, including:

- SEM-EDS spectrum imaging data (hyperspectral elemental mapping)
- Discrete point-based EDS spectra
- FTIR hyperspectral data
- DART-MS measurements
- Supporting metadata and documentation files

This project will focus exclusively on the SEM-EDS spectrum imaging datasets, which provide spatially resolved chemical information suitable for unsupervised clustering and dimensionality-reduction workflows.

Dataset Size & Format:

The SEM-EDS dataset contains approximately 74 files totaling ~55 MB. The data include:

- Two hyperspectral EDS spectrum imaging datasets (paired .raw.gz and .rpl files) corresponding to aerial and cross-sectional views
- Multiple point-based EDS spectra in open .msa format

- Region-of-interest (ROI) TIFF images indicating spectrum acquisition locations
- Standardless quantification tables in .xlsx format
- Supporting metadata and checksum files

The primary data for this project consist of the spectrum imaging datasets, which are stored in vendor-style hyperspectral formats and can be loaded using HyperSpy for conversion into numerical arrays suitable for machine-learning analysis.

Preprocessing Requirements:

Planned preprocessing steps include:

- Loading .rpl/.raw.gz spectral image files using HyperSpy
- Verification of energy calibration
- Background subtraction and elemental peak integration
- Extraction of elemental intensity maps
- Intensity normalization (e.g., total-count normalization)
- Optional spatial smoothing to reduce noise
- Masking of non-sample regions or low-signal regions
- Standardization of elemental features prior to PCA and clustering

Point spectra and ROI quantification tables may be used qualitatively to interpret cluster chemistry but will not serve as primary inputs for the clustering workflow.

Ethical/Proprietary Considerations:

- The dataset is publicly available and intended for open scientific use.
- It contains no personal or sensitive information.
- Use of the dataset requires proper citation of the dataset and associated publication

Preliminary References

1. Romano, C. P.; Lam, T.; Newsome, G. A.; Taillon, J. A.; Little, N. C.; Tsang, J.-S. Characterization of Zinc Carboxylates on an Oil Paint Test Panel. *Stud. Conserv.* **2019**, 64, 221–237. <https://doi.org/10.1080/00393630.2019.1666467>.
2. Van Vijle, A.; Hacıgüzeller, P.; Van der Snickt, G. Machine Learning for Painting Conservation: A State-of-the-Art Review. *npj Heritage Sci.* **2025**, 13, 437. <https://doi.org/10.1038/s40494-025-01924-3>.
3. DeCost, B.L.; Holm, E. A. A Computer Vision Approach for Automated Analysis and Classification of Microstructural Image Data. *Comput. Mater. Sci.* **2015**, 110, 126–133. <https://doi.org/10.1016/j.commatsci.2015.08.011>